

06 — The SPDE Approach (the PDE Duality)

What you'll learn: the exact sense in which a Matérn-covariance GP **is** the solution of a second-order stochastic PDE, why that operator is the place to inject physics, and how the finite-element/GMRF construction turns a dense covariance into a **sparse precision** matrix.

Prereqs: 01, 02;

05 gives the temporal analogue.

Why it matters here: this is the duality you asked about — the route to physical processes, boundary conditions, and cheap compute, all from one equation.

1. The duality in one equation

Whittle (1963) and Lindgren-Rue-Lindström (2011) showed: a Gaussian field with **Matérn covariance**

is the **stationary solution** of the linear **stochastic partial differential equation (SPDE)**

$$(\kappa^2 - \Delta)^{\alpha/2} X(s) = \eta(s)$$

- Δ is the **Laplacian** ($\sum \partial^2 / \partial s_d^2$) — the second-order differential operator. The name "duality

of the second-order PDE" points here.

- $\kappa > 0$ sets the **spatial range** ($\text{range} \approx \sqrt{(8\nu)/\kappa}$); larger $\kappa \rightarrow$ shorter correlations.
- $\alpha = \nu + d/2$ ties the SPDE order to the Matérn smoothness ν and the spatial dimension d .
- η is **Gaussian white noise** (the random forcing).

Read it as: "a Matérn GP is what you get when you pass white noise through the operator $(\kappa^2 - \Delta)^{-\alpha/2}$." The covariance function and the differential operator are two descriptions of the

same object — the duality. (This is the spatial sibling of doc 05:

there a Matérn GP in time equaled an ordinary SDE; here a Matérn GP in space equals a partial SDE.)

2. Why this is powerful — three doors from one equation

Because the GP is now defined by an **operator equation** rather than a fixed covariance formula, you

can edit the operator to change the model's assumptions:

- 1. Inject physical processes.** Replace/augment $(\kappa^2 - \Delta)$ with operators encoding real dynamics —

advection (transport by currents) and **anisotropic diffusion** (mixing). The resulting field automatically has the spatial/temporal correlation structure those physics imply. (Detailed in doc 07.)

- 2. Impose boundary conditions.** A PDE is only well-posed with boundary conditions. Specifying

Dirichlet/Neumann/Robin conditions on the domain boundary makes the GP respect walls, the

seabed, the surface, or known values — exactly, not approximately. (Also doc 07.)

3. Compute cheaply. Solving the SPDE by the **finite element method (FEM)** yields a Gaussian

Markov Random Field (GMRF) **with a** sparse precision matrix — the compute win, below.

3. From SPDE to GMRF: the sparse-precision win

The construction (LRL 2011):

1. Mesh the domain. Triangulate the region of interest (e.g. the operating area / water column)

into nodes and elements.

2. Basis expansion. Approximate the field as $X(s) \approx \sum_k w_k \phi_k(s)$ with local "tent" basis

functions ϕ_k (each nonzero only near node k) and random weights w .

3. Weak form / FEM assembly. Projecting the SPDE onto this basis gives the weights a Gaussian

distribution $w \sim N(0, Q^{-1})$ whose **precision matrix** Q is built from sparse FEM matrices (mass C and stiffness G), e.g. for $\alpha=2$: $Q = (\kappa^2 C + G) C^{-1} (\kappa^2 C + G)$.

Because each basis function overlaps only its neighbours, Q is **sparse** (mostly zeros). This is

the **Markov property in space**: a node is conditionally independent of the rest given its neighbours — the spatial analogue of the block-tridiagonal precision in doc 05 §3.

Dense covariance vs sparse precision — the cost table

	Covariance $\Sigma = K$ (classic GP)	Precision $Q = \Sigma^{-1}$ (GMRF/SPDE)
Structure	<code>**dense**</code> (every point covaries with every point)	<code>**sparse**</code> (only neighbours)
Storage	$O(n^2)$	$O(n)$
Factorize / solve	$O(n^3)$	$\approx O(n^{3/2})$ in 2-D, $O(n)$ in 1-D, using sparse Cholesky
Interpretation	"how outputs correlate"	"conditional dependence graph"

This is why the SPDE/GMRF approach scales to large spatial fields where a classic dense GP cannot — and why it is a serious **edge-compute** option (doc 08).

4. Non-stationarity for free

Let $\kappa = \kappa(s)$ and the diffusion vary over space, and the SPDE produces a **non-stationary** field —

different correlation length/strength in different regions — while keeping the sparse precision. Classic stationary kernels can't do this; encoding it in covariance form is hard, but in operator form

it's just spatially-varying coefficients. Underwater this matters: correlation structure near a seamount differs from open water.

5. Where it does and doesn't fit our project

- **Not the first build.** Our first GP is a temporal DVL-velocity model (doc 04); it doesn't need a spatial mesh.

- **The natural next step.** Modelling the **ocean-current field** the vehicle moves through is inherently spatial (and spatio-temporal). The SPDE/GMRF approach is the right tool: it can encode current physics (advection-diffusion), respect boundaries (seabed/surface), stay non-stationary, and compute on a sparse precision suitable for on-board use. This is item (2) of the edge/SPDE roadmap in implementation/04.
- **Spatio-temporal = combine docs 05 + 06.** Keep Markov-in-time (Kalman) and GMRF-in-space → an efficient current-field filter (Särkkä/Solin infinite-dimensional Kalman; Clarotto et al. for the advection-diffusion SPDE specifically).

6. Tooling note

The classic implementations live in the R-INLA ecosystem (R-INLA, inlabru) and fmesher; Python options include findiff/FEM libraries plus sparse solvers in scipy.sparse. For our roadmap this would be a separate component from the GPyTorch velocity model, justified only when we tackle the current field. Adding it implies a scipy-class dependency for sparse linear algebra and a meshing step.

Key takeaways

- **The duality:** a Matérn GP is the solution of $(\kappa^2 - \Delta)^{\alpha/2} X = \text{white noise}$. Covariance and second-order PDE operator are the same object described two ways.
- Editing the **operator** is how you inject **physics** and impose **boundary conditions** (see doc 07).
- Solving by FEM yields a **GMRF with a sparse precision matrix** → $O(n)$ -ish compute instead of $O(n^3)$, plus easy **non-stationarity**.
- For us: not the first build, but the principled, scalable way to model the **ocean-current field** later.

Further reading

- Lindgren, Rue, Lindström (2011), *An explicit link between Gaussian fields and Gaussian Markov random fields: the SPDE approach*, JRSS-B — **the paper**.
- Whittle (1963) — the original Matérn ↔ SPDE observation.
- Bakka et al. (2018), Spatial modelling with INLA: a review — accessible overview.
- Next: 07 — Physics-informed GPs and boundary conditions.